Module 9: Clustering

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Unsupervised Learning

Unsupervised learning

- We have so far focused on supervised learning methods such as regression and classification.
- In that setting we observe both a set of features $X_1, \ldots, X_p$ for each subject, as well as a response or outcome variable Y. The goal is then to predict Y using $X_1, \ldots, X_p$.
- Here we instead focus on unsupervised learning, we where observe only the features $X_1, \ldots, X_p$. We are not interested in prediction, because we do not have an associated response variable Y.

The goal is to discover interesting things about the measurements: is there an informative way to visualize the data? Can we discover subgroups among the variables or among the observations?

We first discuss clustering, a broad class of methods for discovering unknown subgroups in data.

The challenge of unsupervised learning

- Unsupervised learning is more subjective than supervised learning, as there is no simple goal for the analysis, such as prediction of a response.
- But techniques for unsupervised learning are of growing importance in a number of fields:
  - subgroups of breast cancer patients grouped by their gene expression measurements,
  - groups of shoppers characterized by their browsing and purchase histories,
  - movies grouped by the ratings assigned by movie viewers.

Another advantage

- It is often easier to obtain unlabeled data — from a lab instrument or a computer — than labeled data, which can require human intervention.
- For example it is difficult to automatically assess the overall sentiment of a movie review: is it favorable or not?

Unsupervised learning

Training sample: x_i; $i = 1, \ldots, n$ with $x_i \in \mathbb{R}^p$, and the goal is to infer properties of $Pr(X)$ without the help of responses.

- Density estimation: to estimate $Pr(x)$ directly; well studied in statistical literature, and not a focus of this lecture
- Cluster analysis: to partition training sample into a number of subgroups such that observations in the same subgroup are similar
- Manifold learning: to learn latent structure and to visualize data in low dimensions

K-Means

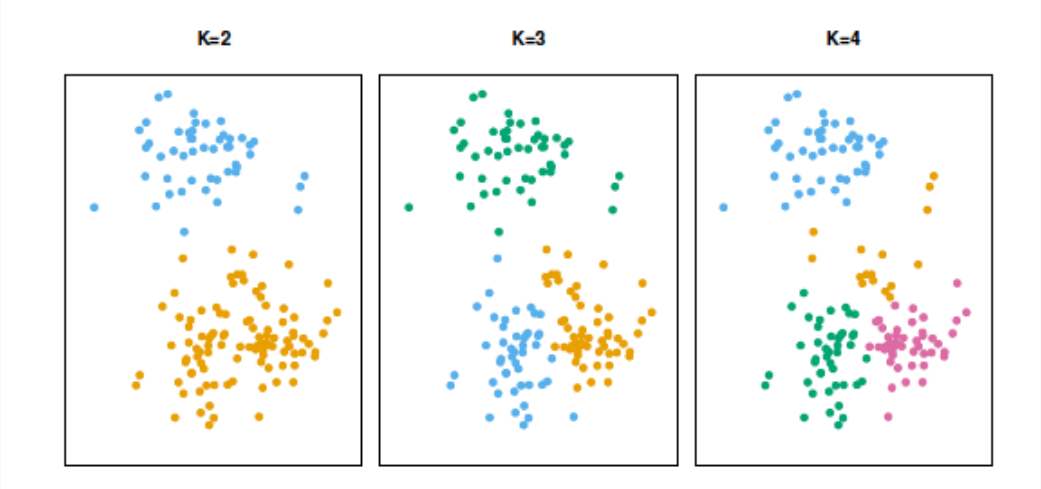
A motivation example: Clustering for market segmentation

- Suppose we have access to a large number of measurements (e.g. median household income, occupation, distance from nearest urban area, and so forth) for a large number of people.
- Our goal is to perform market segmentation by identifying subgroups of people who might be more receptive to a particular form of advertising, or more likely to purchase a particular product.
- The task of performing market segmentation amounts to clustering the people in the data set.

Two clustering methods

- In K-means clustering, we seek to partition the observations into a pre-specified number of clusters.
- In hierarchical clustering, we do not know in advance how many clusters we want; in fact, we end up with a tree-like visual representation of the observations, called a dendrogram, that allows us to view at once the clusterings obtained for each possible number of clusters, from 1 to n.

A simulated example of clustering



A simulated data set with 150 observations in 2-dimensional space. Panels show the results of applying K-means clustering with different values of K, the number of clusters. The color of each observation indicates the cluster to which it was assigned using the K-means clustering algorithm. Note that there is no ordering of the clusters, so the cluster coloring is arbitrary. These cluster labels were not used in clustering; instead, they are the outputs of the clustering procedure.

Dissimilarity matrix

To conduct cluster analysis, we first need to define a dissimilarity measure $d(x_i, x_{i'})$.

- The most commonly used measure is Euclidean distance $d(x_i, x_{i'}) = \|x_i - x_{i'}\|^2$
- Other measures are possible:
  - Absolute error: $d(x_i, x_{i'}) = \|x_i - x_{i'}\|_1$
  - Correlation: $d(x_i, x_{i'}) = \rho(x_i, x_{i'})$
  - For nominal variables, $d(x_i, x_{i'}) = \sum_{j=1}^{p} I(x_{ij} \neq x_{i'j})$
- Weights can be used to adjust importance of covariates
- A dissimilarity matrix $\mathbf{D} = (d(x_i, x_{i'}))_{i,i'=1}^{n}$

Combinatorial clustering

Assume the number of cluster is pre-specified as K, and $a(x) : \mathbb{R}^p \to \{1, \ldots, K\}$ is a clustering assignment function.

A clustering algorithm is to construct $a(x)$ to minimize

$$W(a) = \sum_{k=1}^{K} \sum_{a(x_i)=k} \sum_{a(x_{i'})=k} d(x_i, x_{i'}),$$

which can be regarded as "within-cluster variability".

Analogously,

$$T = \sum_{i=1}^{n} \sum_{i'=1}^{n} d(x_i, x_{i'}),$$

$$B(a) = \sum_{k=1}^{K} \sum_{a(x_i)=k} \sum_{a(x_{i'})\neq k} d(x_i, x_{i'}),$$

which are regarded as total variability and between-cluster variability. Clearly,

$$T = B(a) + W(a),$$

so minimizing $W(a)$ is the same as maximizing $B(a)$.

The optimization is a combinatorial optimization problem, and it is usually intractable.

K-means clustering

Assume there are K centers, $c_1, \ldots, c_K$, then x_i is assigned to the $a(x_i)$-th center if

$$a(x_i) = \operatorname*{argmin}_k \|x_i - c_k\|^2.$$

The goal of K-means is to find $c_1, \ldots, c_K$ so that the total mean squared error between the training samples and their corresponding centers is minimized. In specific,

$$\operatorname*{argmin}_{c_1,\ldots,c_K} \sum_{i=1}^{n} \|x_i - c_{a(x_i)}\|^2.$$

K-means clustering algorithm

1. Randomly assign a number, from 1 to K, to each of the observations. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
   - 2.1 For each of the K clusters, compute the cluster centroid. The kth cluster centroid is the vector of the p feature means for the observations in the kth cluster.
   - 2.2 Assign each observation to the cluster whose centroid is closest (where closest is defined using Euclidean distance).

K-means clustering algorithm

An iterative greedy descent scheme can be employed.

- If $c_1, \ldots, c_K$ are fixed, the assignment function $a(\cdot)$ is

$$a(x_i) = \underset{k}{\operatorname{argmin}} \|x_i - c_k\|^2$$

- If $a(x_i)$'s are fixed, then

$$c_k = \frac{\sum_{i=1}^{n} x_i I(a(x_i) = k)}{\sum_{i=1}^{n} I(a(x_i) = k)}$$

- Iterate the above two steps until convergence

Remarks on K-means clustering

- The algorithm always converges as the objective function is non-increasing along the iteration
- But the convergence point may be a suboptimal local minimizer, and most times it is
- And the convergence point is highly dependent on the initial value
- To overcome the variability, random starts are often used, and choose the one having smallest objective function value

Properties of K-means algorithm

- Guaranteed to converge in a finite number of iterations
- Running time per iteration:
  1. Assign data points to closest cluster center $O(KN)$ time
  2. Change the cluster center to the average of its assigned points $O(N)$

Example

- Training sample: $1.2, 5.6, 3.7, 0.6, 0.1, 2.6$
- Apply K-means clustering with $K = 2$
- Initialization: $c_1 = 2,\ c_2 = 5$

step	fixed	update
1	$c_1 = 1,\ c_2 = 5$	$\{1.2, 0.6, 0.1, 2.6\}, \{5.6, 3.7\}$
2	$\{1.2, 0.6, 0.1, 2.6\}, \{5.6, 3.7\}$	$c_1 = 1.125,\ c_2 = 4.65$
3	$c_1 = 1.125,\ c_2 = 4.65$	$\{1.2, 0.6, 0.1, 2.6\}, \{5.6, 3.7\}$

- Converged after three steps, yields $c_1 = 1.125,\ c_2 = 4.65$ and the objective function value is 5.3125

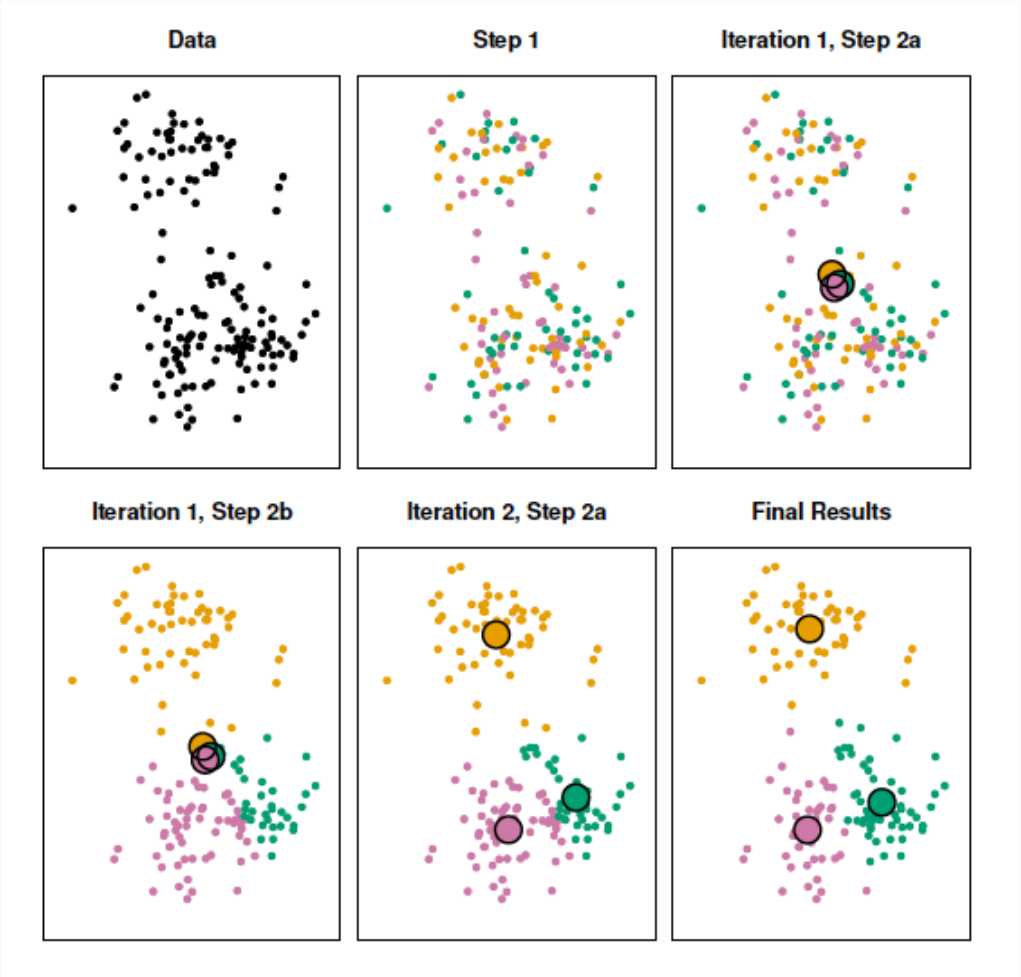
Example: cont'd

- Initialization: $c_1 = 0.8,\ c_2 = 3.8$

step	fixed	update
1	$c_1 = 0.8,\ c_2 = 3.8$	$\{1.2, 0.6, 0.1\}, \{5.6, 3.7, 2.6\}$
2	$\{1.2, 0.6, 0.1\}, \{5.6, 3.7, 2.6\}$	$c_1 = 0.633,\ c_2 = 3.967$
3	$c_1 = 0.633,\ c_2 = 3.967$	$\{1.2, 0.6, 0.1\}, \{5.6, 3.7, 2.6\}$

- Converged after three steps, yields $c_1 = 0.633,\ c_2 = 3.967$ and the objective function value is 5.2133
- It can be shown that $c_1 = 0.633,\ c_2 = 3.967$ is the global minimizer

Example


Data
Step 1
Iteration 1, Step 2a
Iteration 1, Step 2b
Iteration 2, Step 2a
Final Results

Details of previous figure

The progress of the K-means algorithm with K=3.

- Top left: The observations are shown.
- Top center: In Step 1 of the algorithm, each observation is randomly assigned to a cluster.
- Top right: In Step 2(a), the cluster centroids are computed. These are shown as large colored disks. Initially the centroids are almost completely overlapping because the initial cluster assignments were chosen at random.
- Bottom left: In Step 2(b), each observation is assigned to the nearest centroid.
- Bottom center: Step 2(a) is once again performed, leading to new cluster centroids.
- Bottom right: The results obtained after 10 iterations.

Example: different starting values



Details of previous figure

K-means clustering performed six times on the data from previous figure with $K = 3$, each time with a different random assignment of the observations in Step 1 of the K-means algorithm.

Above each plot is the value of the objective function. Three different local optima were obtained, one of which resulted in a smaller value of the objective and provides better separation between the clusters. Those labeled in red all achieved the same best solution, with an objective value of 235.8

K-medoids clustering

Similar to K-means clustering, but with a few modifications:

- The Euclidean distance is replaced by any distance measure $d(x_i, x_{i'})$
- The cluster centers are restricted to be one of the sample points $c_k = x_{i_k^*}$, where

$$i_k^* = \underset{\{i:a(x_i)=k\}}{\operatorname{argmin}} \sum_{a(x_{i'})=k} d(x_i, x_{i'})$$

- Can take dissimilarity matrix $\mathbf{D}$ as input
- Far more computationally intensive than K-means

K-means++ Initialization: D^2 sampling

- Let $D(x)$ be the distance between a point x and its nearest center. Chose the next center proportional to $D^2(x)$.
- Choose c_1 as random
  For $j = 2, \ldots, K$, pick c_j among $x_1, \ldots, x_n$ according to this distribution

$$P(c_j = x_i) \propto \min_{j' < j} \|x_i - c_{j'}\|^2$$

- Theorem: K-means++ always attains an $O(\log K)$ approximation to optimal k-means solution in expectation.
- Running the general K-means can only further improve the cost.

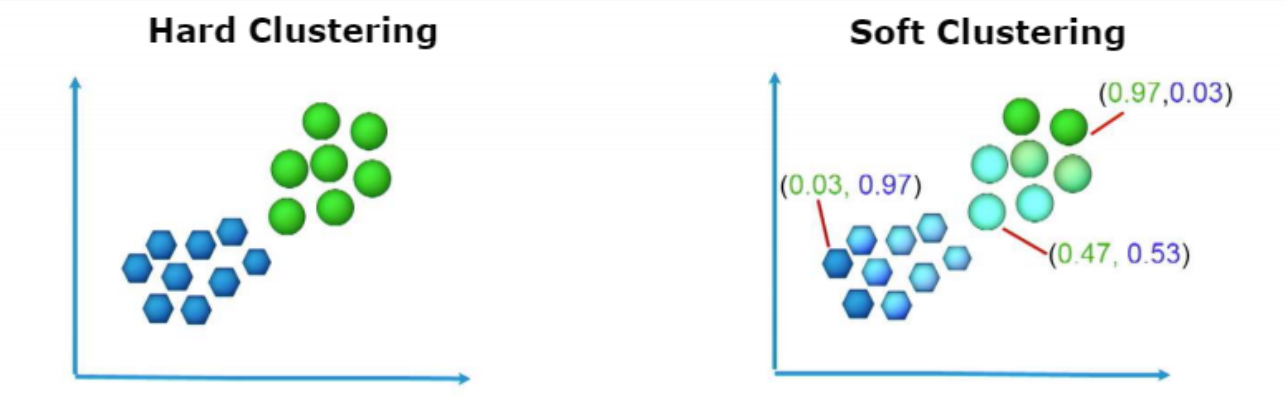
- Interpolate between random and furthest point initialization
- Let $D(x)$ be the distance between a point x and its nearest center. Chose the next center proportional to $D(x)^{\alpha}$:
  — $\alpha = 0$: random sampling
  — $\alpha = \infty$: furthest point
  — $\alpha = 2$: K-means++

What value of k?

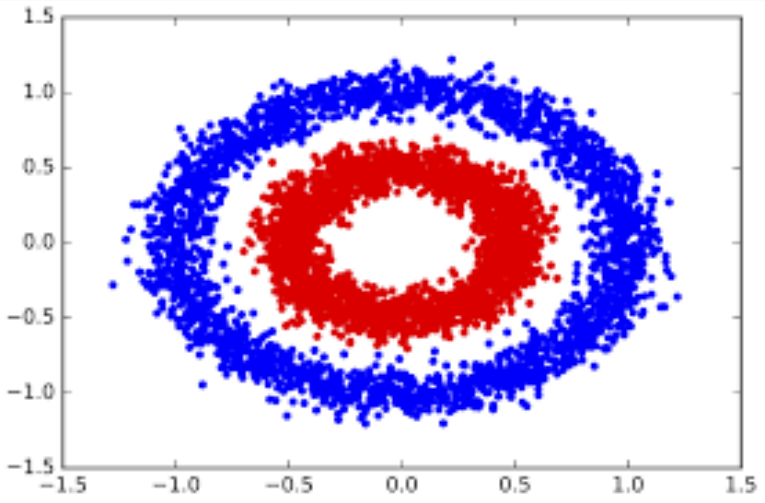
- Hold-out validation/cross-validation on auxiliary task (e.g., supervised learning task).
- Heuristic: Find large gap between $k-1$-means cost and k-means cost.
- More details are coming.
- Try hierarchical clustering.

Later...

Soft K-means



Kernel K-means



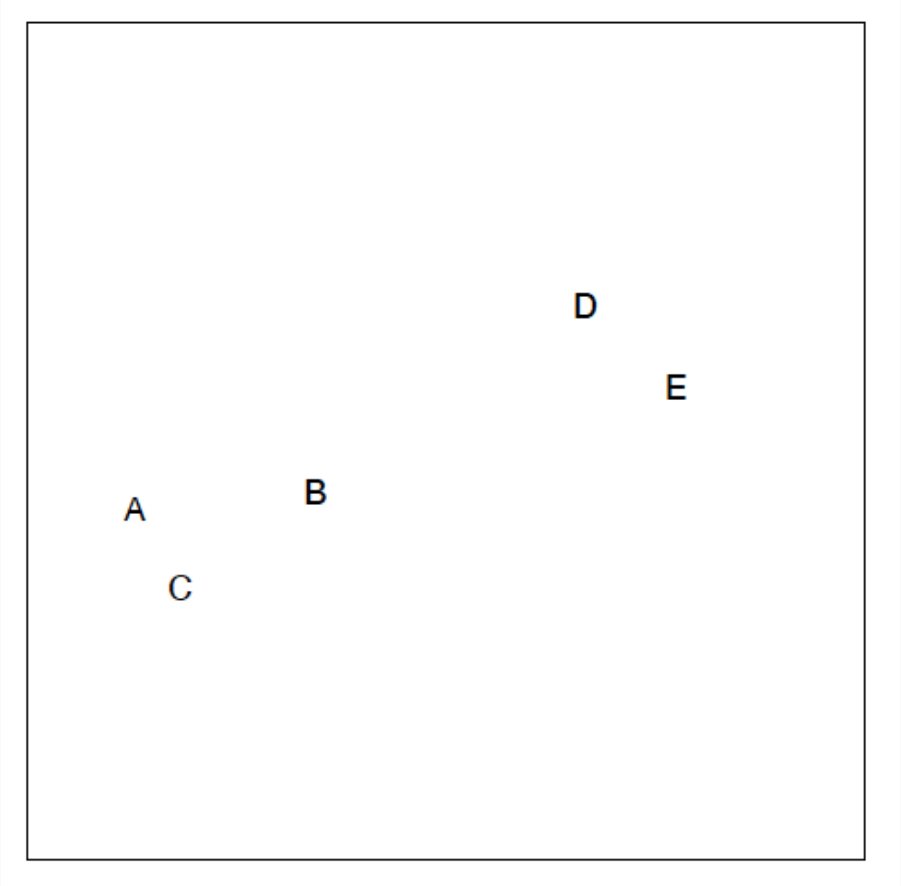
Hierarchical Clustering

Hierarchical clustering

- K-means clustering requires us to pre-specify the number of clusters K. This can be a disadvantage (later we discuss strategies for choosing K)
- Hierarchical clustering is an alternative approach which does not require that we commit to a particular choice of K.
- In this section, we describe bottom-up or agglomerative clustering. This is the most common type of hierarchical clustering, and refers to the fact that a dendrogram is built starting from the leaves and combining clusters up to the trunk.

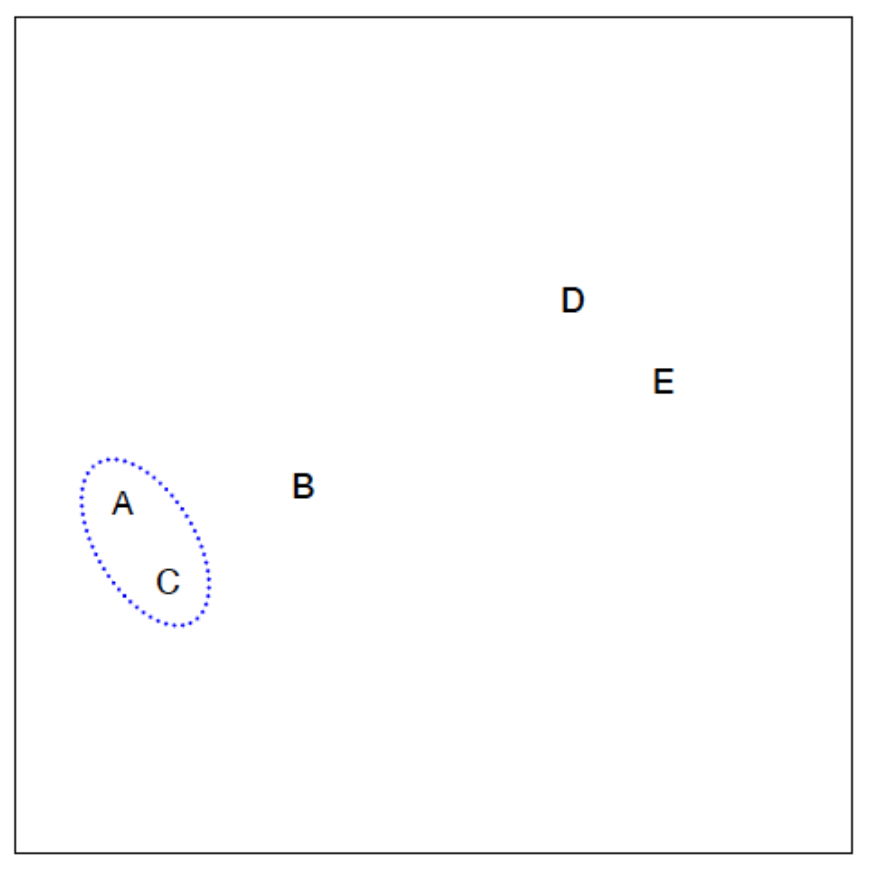
Hierarchical clustering: the idea

Builds a hierarchy in a "bottom-up" fashion...



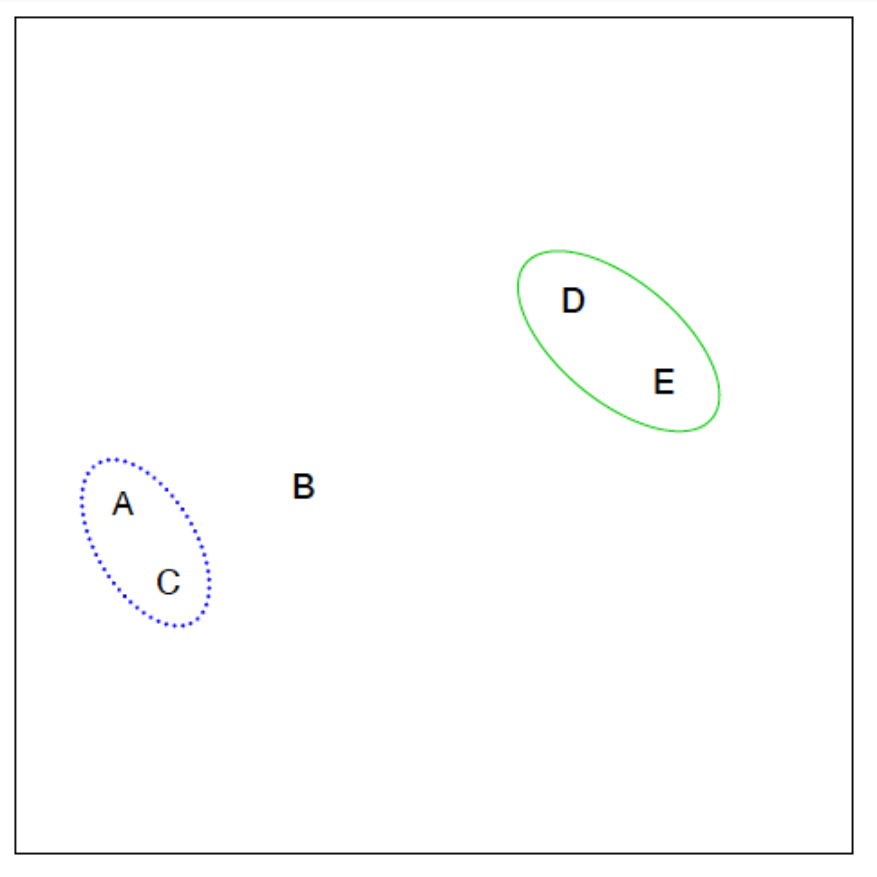
Hierarchical clustering: the idea

Builds a hierarchy in a "bottom-up" fashion...



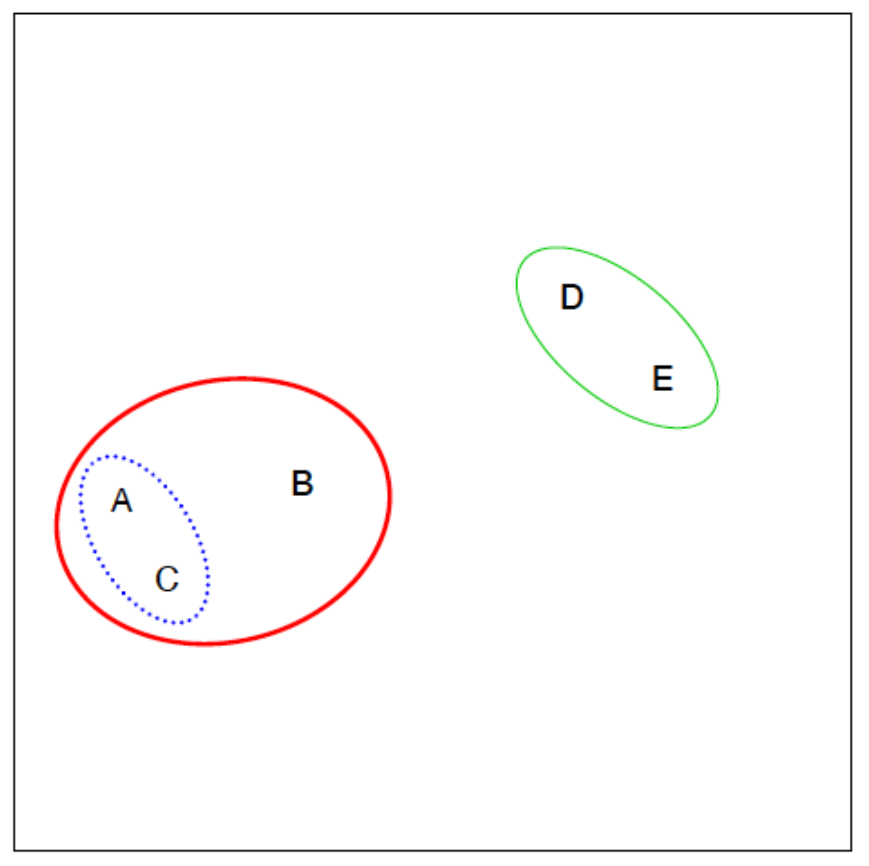
Hierarchical clustering: the idea

Builds a hierarchy in a "bottom-up" fashion...



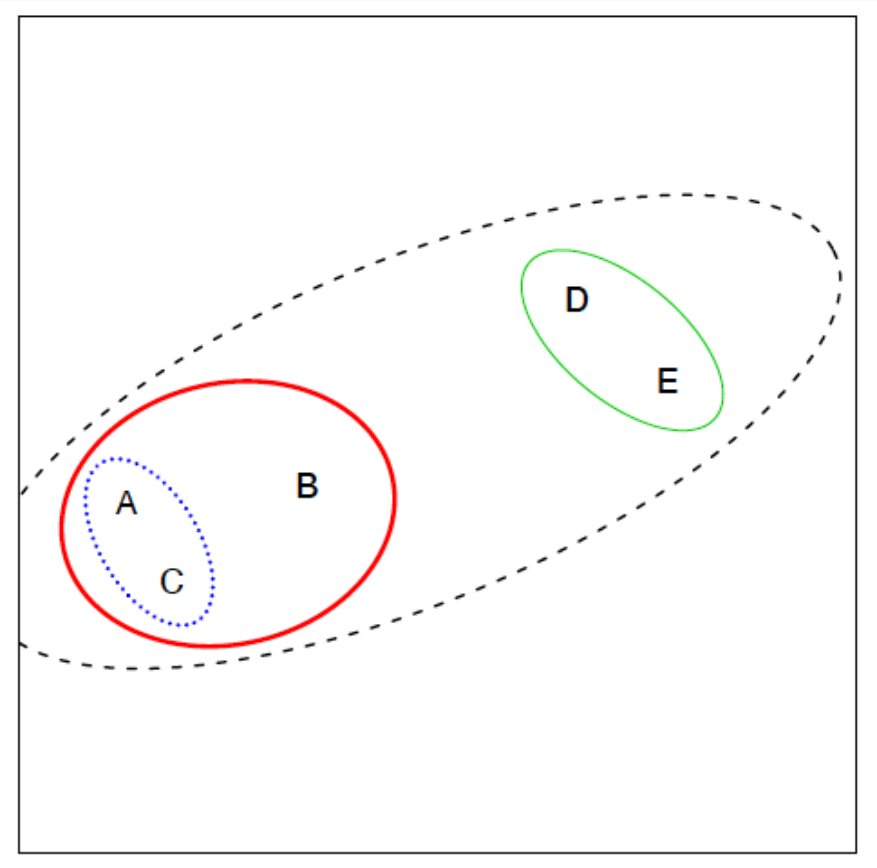
Hierarchical clustering: the idea

Builds a hierarchy in a "bottom-up" fashion...



Hierarchical clustering: the idea

Builds a hierarchy in a "bottom-up" fashion...

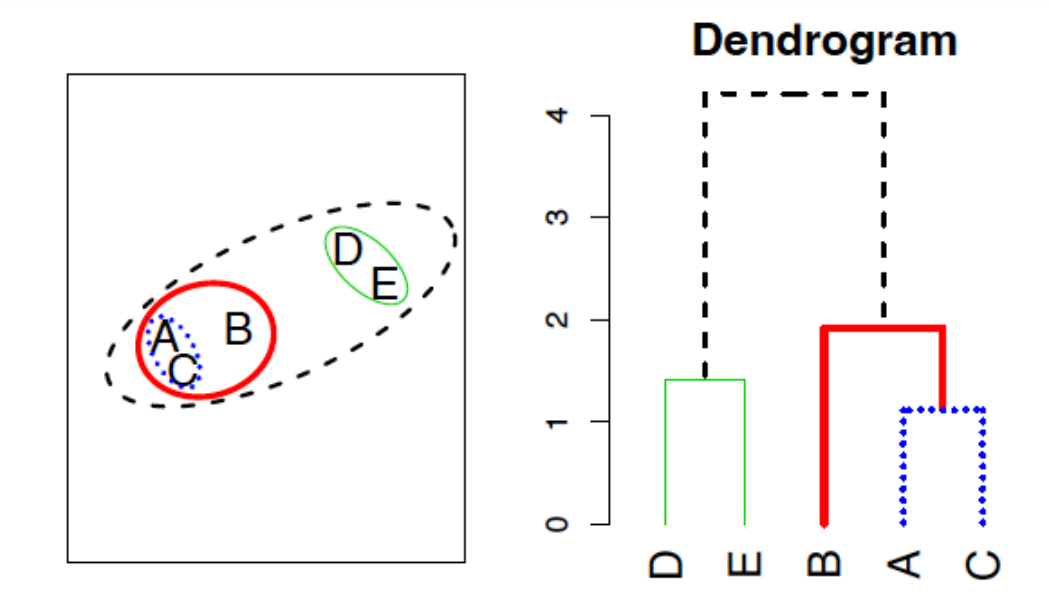


Hierarchical clustering algorithm

Agglomerative clustering:

- Generate clusters in a hierarchical way
- Start with n clusters, each containing 1 data point
- Merge the two clusters with minimum between-cluster dissimilarity
- Update between-cluster distance
- Iterate the merging procedure until all points are in the same cluster
- The clustering procedure can be visualized by a tree structure called dendrogram

Hierarchical clustering algorithm



Hierarchical clustering

Divisive clustering:

- Start with 1 cluster containing all data points
- Split one of the existing clusters into two new clusters with the largest between-cluster dissimilarity
- Update between-cluster distance
- Iterate the splitting procedure until each cluster contains 1 data point

Both agglomerative clustering and divisive clustering possess monotonicity in between-cluster dissimilarity.

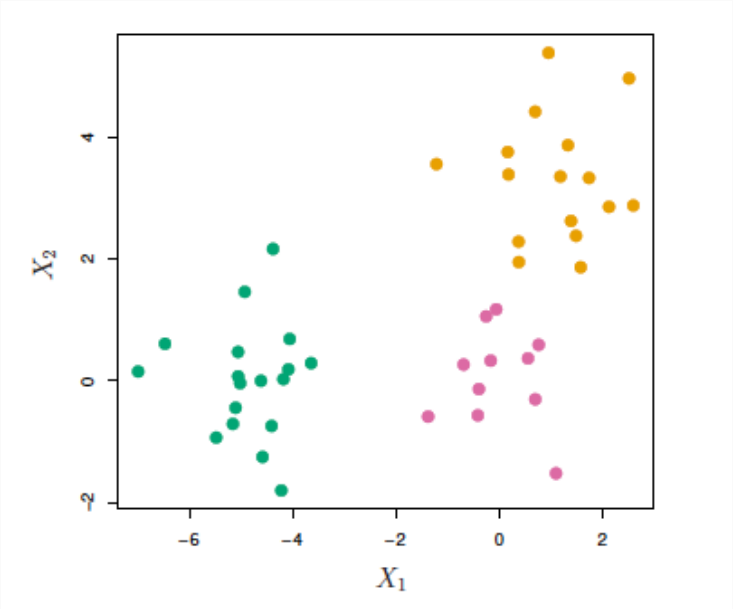
Agglomerative clustering

Definition for between-cluster dissimilarity

- For clusters G and H containing only one data point, the between-cluster dissimilarity $d(G, H)$ is the dissimilarity between two points
- For clusters G and H containing multiple data points, the between-cluster dissimilarity $d(G, H)$ is an agglomerative version of the between-point dissimilarities
  - Single linkage: $d_{sl}(G, H) = \min_{i \in G, i' \in H} d(x_i, x_{i'})$
  - Complete linkage: $d_{cl}(G, H) = \max_{i \in G, i' \in H} d(x_i, x_{i'})$
  - Group average: $d_{ga}(G, H) = |G|^{-1}|H|^{-1} \sum_{i \in G, i' \in H} d(x_i, x_{i'})$
  - Centroid: $d_{cen}(G, H) = d(\bar{x}_G, \bar{x}_H)$

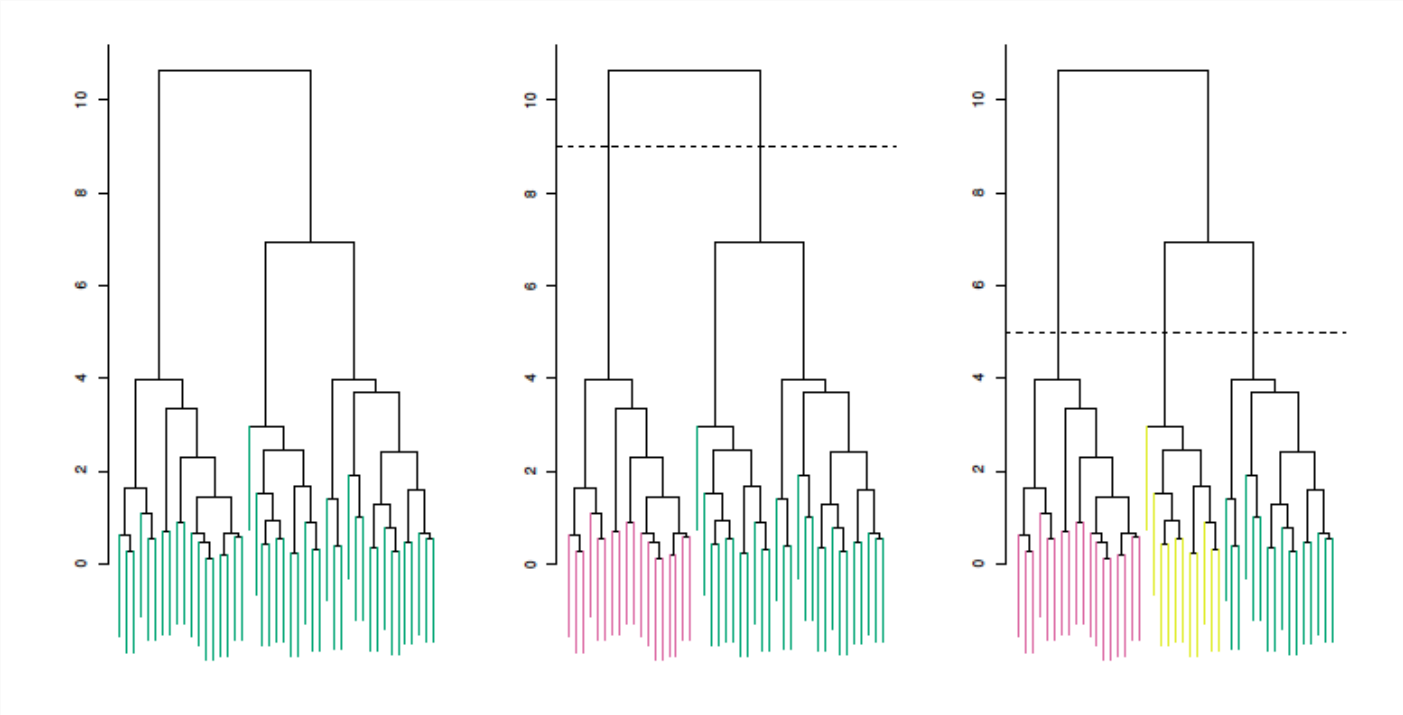
- Complete: Maximal inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the largest of these dissimilarities.
- Single: Minimal inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the smallest of these dissimilarities.
- Average: Mean inter-cluster dissimilarity. Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the average of these dissimilarities.
- Centroid: Dissimilarity between the centroid for cluster A (a mean vector of length p) and the centroid for cluster B.

An example



45 observations generated in 2-dimensional space. In reality there are three distinct classes, shown in separate colors. However, we will treat these class labels as unknown and will seek to cluster the observations in order to discover the classes from the data.

Application of hierarchical clustering



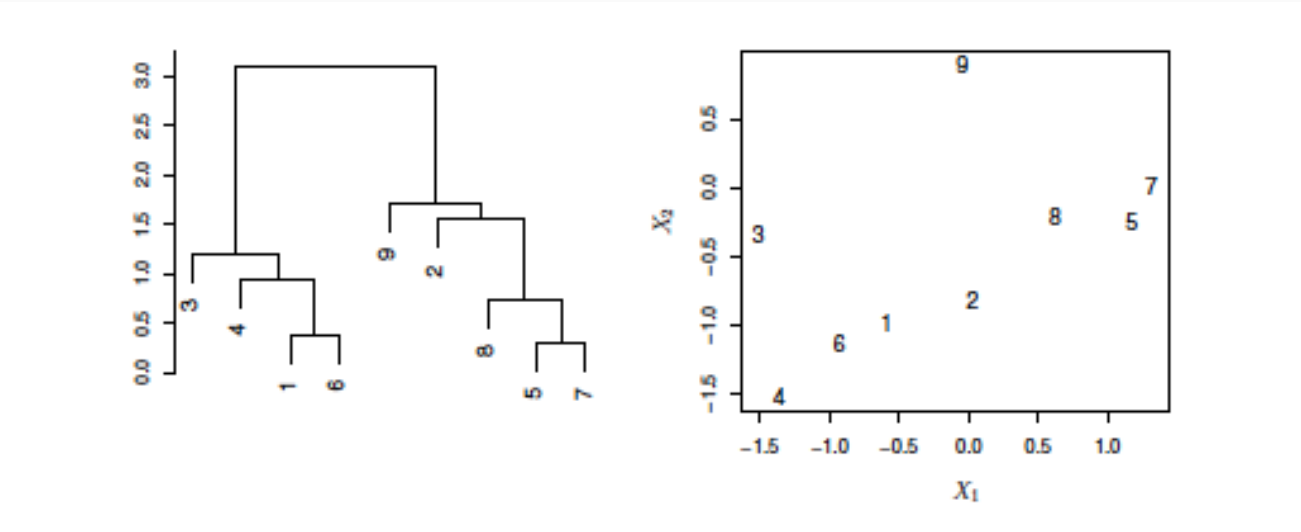
What exactly does the y-axis "Height" mean?

— The horizontal axis represents the clusters. The vertical scale on the dendrogram represent the distance or dissimilarity. Each joining (fusion) of two clusters is represented on the diagram by the splitting of a vertical line into two vertical lines. The vertical position of the split, shown by a short bar gives the distance (dissimilarity) between the two clusters.

Details of previous figure

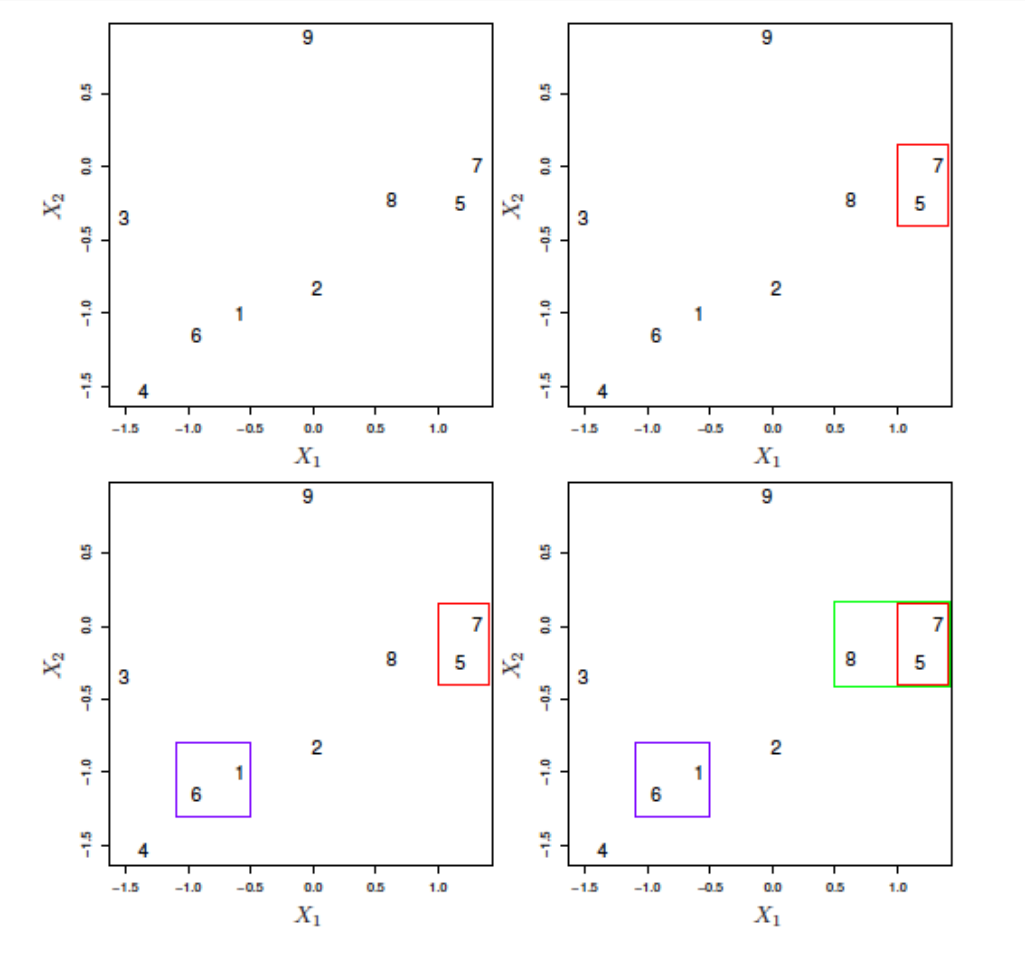
- Left: Dendrogram obtained from hierarchically clustering the data from previous slide, with complete linkage and Euclidean distance.
- Center: The dendrogram from the left-hand panel, cut at a height of 9 (indicated by the dashed line). This cut results in two distinct clusters, shown in different colors.
- Right: The dendrogram from the left-hand panel, now cut at a height of 5. This cut results in three distinct clusters, shown in different colors. Note that the colors were not used in clustering, but are simply used for display purposes in this figure

Another example



- An illustration of how to properly interpret a dendrogram with nine observations in two-dimensional space. The raw data on the right was used to generate the dendrogram on the left.
- Observations 5 and 7 are quite similar to each other, as are observations 1 and 6.
- However, observation 9 is no more similar to observation 2 than it is to observations 8, 5, and 7, even though observations 9 and 2 are close together in terms of horizontal distance.
- This is because observations 2, 8, 5, and 7 all fuse with observation 9 at the same height, approximately 1.8.

Merges in previous example



Scaling of the variables matters

Incomparable units. Height vs weight. You cannot compare, so the default decision is to standardize (equalize variances); it is "default" on the grounds of thought parsimony: "every unique aspect of nature is assumed to have same, unit variability of observations".

Same units, irrelative features. Height vs circumference. These are clearly independent (conceptually, not statistically) phenomena of reality. Their same-unitness seems a coincidence. It would be silly to compare between the two values. The default decision is to standardize the features.

Same units, juxtaposed features. Length of right arm vs of left arm. We could naturally compare the two lengths if we need so, they two are interchangeable, in a sense. The default decision is to leave variances as is (no matter how much they differ). Because "leave nature under study be how it is".

Undecided whether 2 or 3. Length of arm vs length of leg. We could compare these but we are not interested in that, rather, we prefer to see the lengths as separate dimensions (albeit not irrelative phenomena). Feature-conceptual decision (whether standardize or leave) is impossible.

Practical issues

- Should the observations or features first be standardized in some way? For instance, maybe the variables should be centered to have mean zero and scaled to have standard deviation one.
- In the case of hierarchical clustering,
  — What dissimilarity measure should be used?
  — What type of linkage should be used?
- How many clusters to choose? (in both K-means or hierarchical clustering). Difficult problem. No agreed-upon method. See Elements of Statistical Learning, chapter 13 for more details.

More on the between-cluster similarity

- Single linkage tends to cause *chaining* and thus violate the "compactness" property
  — It will therefore have a tendency to combine, at relatively low thresholds, observations linked by a series of close intermediate observations. This phenomenon, referred to as chaining, is often considered a defect of the method.
  — The clusters produced by single linkage can violate the "compactness" property that all observations within each cluster tend to be similar to one another, based on the supplied observation dissimilarities $\{d_{ii}\}$.

- Complete linkage tends to violate the "closeness" property
  — Two groups G and H are considered close only if all of the observations in their union are relatively similar. It will tend to produce compact clusters with small diameters. However, it can produce clusters that violate the "closeness" property. That is, observations assigned to a cluster can be much closer to members of other clusters than they are to some members of their own cluster.
- Group average compromises the single linkage and the complete linkage, but it is not invariant under a monotone transformation

Each cluster k is a random sample from some population joint density $p_k(x)$. The complete data set is a random sample from a mixture of K such densities.

- As n increases, group average approaches

$$\int d(x, x')p_G(x)p_H(x')dxdx',$$

  but single linkage approaches 0 and complete linkage approaches ∞

Thus, it is not clear what aspects of the population distribution are being estimated by $d_{sl}(G, H)$ and $d_{cl}(G, H)$.

Example: microarray data

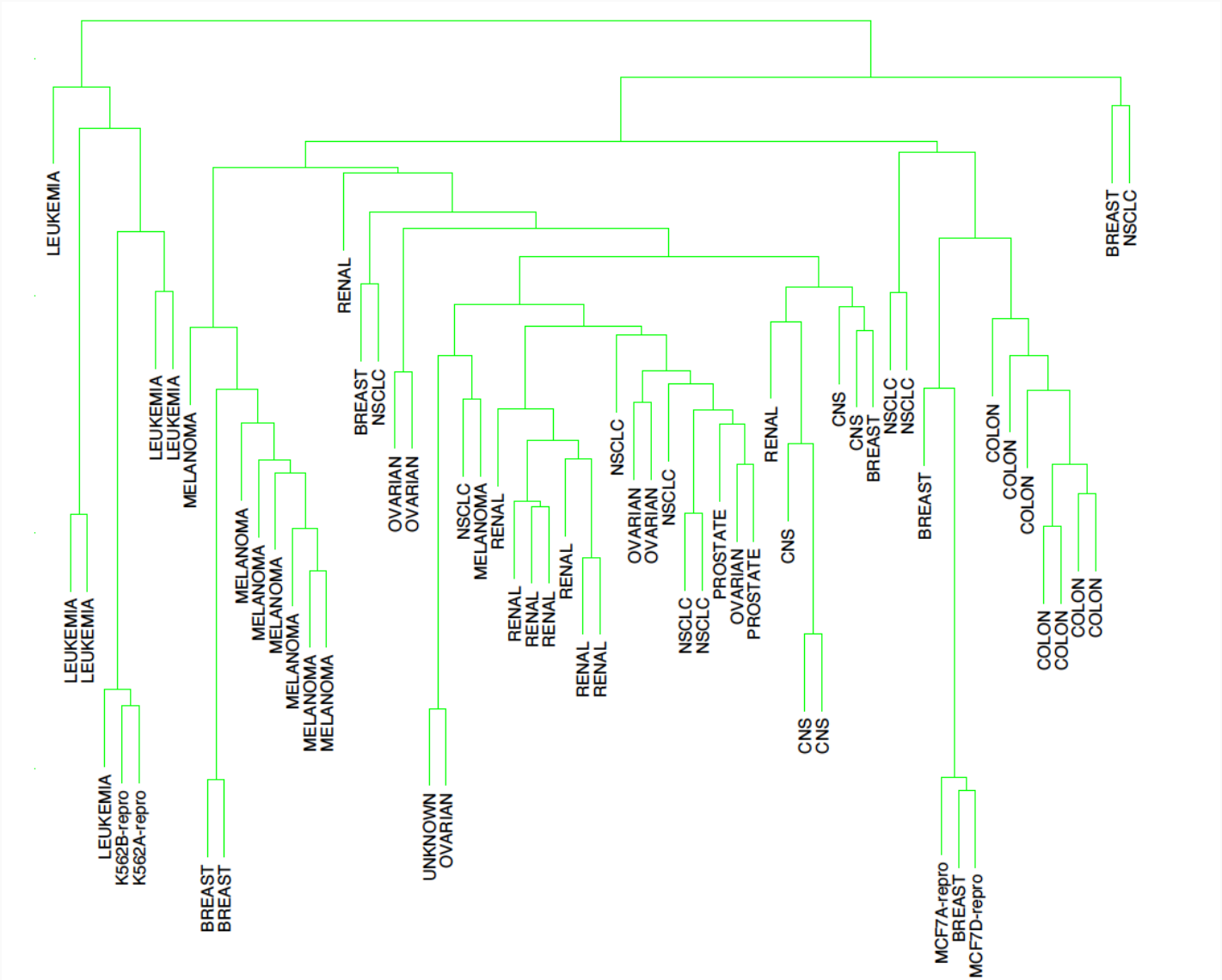


FIGURE 14.12. *Dendrogram from agglomerative hierarchical clustering with average linkage to the human tumor microarray data.*

Model-Based Clustering

Model-based clustering

A popular model-based clustering is based on Gaussian mixture model (GMM);

$$X \sim \sum_{k=1}^{K} \alpha_k \phi(x; \mu_k, \Sigma_k)$$

- α_k is mixing proportion with $\sum_k \alpha_k = 1$
- $\phi(x; \mu_k, \Sigma_k)$ is a Gaussian density for each mixing component
- Other density components are possible, but Gaussian density is most popular
- Introducing $G \in \{1, \ldots, K\}$, then GMM is equivalent to

$$P(G = k) = \alpha_k, \quad X|G = k \sim \phi_k(x; \mu_k, \Sigma_k),$$

and the goal of clustering is $P(G = k|X = x)$

Let $G_{ik} \in \{0, 1\}$ indicate cluster membership. As it is missing, GMM can be estimated by the EM algorithm:

- Let $\theta_k = (\mu_k, \Sigma_k)$
- E step: $G_{ik}^{(t)} = \frac{\alpha_k^{(t)} \phi_k(x; \theta_k^{(t)})}{\sum_k \alpha_k^{(t)} \phi_k(x; \theta_k^{(t)})}$, and then

$$Q(\theta, \theta^{(t)}) = \sum_{i=1}^{n} \sum_{k=1}^{K} G_{ik}^{(t)} \big(\log \alpha_k + \log \phi_k(x; \theta_k) \big)$$

- M step: $\alpha_k^{(t+1)} = \frac{\sum_{i=1}^n G_{ik}^{(t)}}{n}$,

$$\begin{aligned} \mu_k^{(t+1)} &= \frac{\sum_{i=1}^n G_{ik}^{(t)} x_i}{\sum_{i=1}^n G_{ik}^{(t)}} \\ \Sigma_k^{(t+1)} &= \frac{\sum_{i=1}^n G_{ik}^{(t)} (x_i - \mu_k^{(t+1)})(x_i - \mu_k^{(t+1)})^T}{\sum_{i=1}^n G_{ik}^{(t)}} \end{aligned}$$

Some remarks

- Model-based clustering allows for mixed cluster membership
- It provides a well-defined target for cluster analysis

- EM is guaranteed to converge
- Random initialization

- When dimension is high, some structure is often given:
  - Sparse mean and covariance matrix
  - Diagonal covariance matrix

A second look at K-means $\rightarrow$ Kernel K-means

Define a cluster assignment matrix $L = (L_{ik})$; $i = 1, \ldots, n$; $k = 1, \ldots, K$, whose entries are either 0 or 1 and each row sum is 1, and $L_{ik} = 1$ means x_i belongs to the k-th cluster. Then

$$C = (L^T L)^{-1} L^T X$$

is the matrix of all center vectors, and K-means clustering solves

$$\min_L \ tr(X^T (I - M)^T (I - M) X),$$

where $M = L(L^T L)^{-1} L^T$, similar as the hat matrix in regression.

The formulation can be simplified to

$$\min_L \ tr(X^T(I-M)X) = \min_L \ tr(XX^T(I-M)),$$

therefore it is equivalent to solve

$$\max_L \ tr(XX^T M) = \max_L \ tr(XX^T L(L^T L)^{-1} L^T).$$

Note that it only relies on XX^T, so we can use kernel tricks, and replace XX^T by

$$h(X)h(X)^T = \mathbf{K} = (K(x_i, x_{i'}))_{i,i'=1}^n.$$

Kernel K-means

$$\max_{L} \; tr(\mathbf{K}\, L(L^T L)^{-1} L^T)$$

Note that $L^T L = \text{diag}(n_1, \ldots, n_K)$ with n_k being the clusters sizes. Denote $N = (L^T L)^{-1/2}$ and $A = LN$, the formulation becomes

$$\max_{A} \; tr(A^T\, \mathbf{K}\, A),$$

where $A^T A = (L^T L)^{-1/2}(L^T L)(L^T L)^{-1/2} = I$. Therefore, the solutions are simply the first K eigenvectors of $\mathbf{K}$.

Spectral clustering

$A = LN$, where L has a single 1 per row and N is diagonal, so A can have only one nonzero entry per row and a total of K different nonzero values.

To estimate L based on A,

- Threshold the largest value in each row of A as 1 and the remaining as 0
- First normalize A so that each row of A is unit length, and then run K-means on the normalized A to construct K clusters (spectral clustering)

Selection of the number of cluster

- Most criteria assess the clustering quality by measuring closeness among observations in the same cluster
  - E.g. Calinski and Harabasz (1974), Hartigan (1975), Kaufman and Rousseeuw (1990), Tibshirani et al. (2001), Sugar and James (2003), and many others
  - Key idea is to measure the between-cluster dissimilarity and adjust to the number of clusters
- A general criteria based on clustering stability

Clustering stability

- Assume $z^n = (x_1, ..., x_n)$ are iid from $p(x)$ with $x \in \mathbb{R}^p$. A clustering $\psi(x)$ is defined as a mapping $\psi : \mathbb{R}^p \to \{1, \ldots, k\}$,

Clustering stability

- Assume $z^n = (x_1, ..., x_n)$ are iid from $p(x)$ with $x \in \mathbb{R}^p$. A clustering $\psi(x)$ is defined as a mapping $\psi : \mathbb{R}^p \to \{1, \ldots, k\}$, and a clustering algorithm $\Psi(\cdot; k)$ with a given number of clusters $k \geq 2$ yields a clustering mapping $\psi(x)$ when applied to a sample z^n.
  - The case of $k = 1$ is excluded as $\Psi(\cdot; 1)$ leads to the same degenerated clustering $\psi(x) \equiv 1$ regardless of z^n.
- Motivation is that if repeatedly draw samples from $p(x)$ and apply the candidate $\Psi(\cdot; k)$, a good $\Psi(\cdot; k)$ should produce clusterings $\psi(x)$ that do not vary much from one sample to another.

Clustering distance

Definition (Clustering distance)
The distance between any two clusterings $\psi_1(x)$ and $\psi_2(x)$ is

$$d(\psi_1, \psi_2) = \text{pr}\Big(I(\psi_1(X) = \psi_1(Y)) + I(\psi_2(X) = \psi_2(Y)) = 1 \Big),$$

where X and Y are independently sampled from $p(x)$.

This is a legitimate distance measure and satisfies

- (Nonnegativity) $d(\psi_1, \psi_2) \geq 0$;
- (Symmetry) $d(\psi_1, \psi_2) = d(\psi_2, \psi_1)$;
- (Triangle inequality) $d(\psi_1, \psi_2) + d(\psi_2, \psi_3) \geq d(\psi_1, \psi_3)$.

Definition (Clustering instability)
The clustering instability of $\Psi(\cdot; k)$ is

$$s(\Psi, k, n) = E\Big(d(\Psi(Z^n; k), \Psi(Z^{*n}; k))\Big),$$

where E is taken w.r.t. Z^n and Z^{*n} that are two independent samples of size n from $p(x)$, and $\Psi(Z^n; k)$ and $\Psi(Z^{*n}; k)$ are clusterings obtained by applying $\Psi(\cdot; k)$ to Z^n and Z^{*n} respectively.

- $0 \leq s(\Psi, k, n) \leq 1$, and small $s(\Psi, k, n)$ indicates a stable clustering algorithm $\Psi(\cdot; k)$.
- Inappropriate number of clusters may lead to large $s(\Psi, k, n)$.
- $s(\Psi, k, n)$ is generally unknown and needs to be estimated.

Leave-many-out cross validation

- The key idea is to split the data into two training sets and one validation set. Two clusterings can be obtained by applying $\Psi(\cdot; k)$ to the two training sets, and their distance is measured on the validation set.
- Repeat the splitting many times, and average the estimated clustering instability.
- Select the number of cluster that yields the smallest averaged clustering instability.

Conclusions

- Unsupervised learning is important for understanding the variation and grouping structure of a set of unlabeled data, and can be a useful pre-processor for supervised learning
- It is intrinsically more difficult than supervised learning because there is no gold standard (like an outcome variable) and no single objective (like test set accuracy)
- It is an active field of research, with many recently developed tools such as VAE and GAN.

Additional readings

Schaeffer, E. (2007). Graph clustering. *Computer Science review*, **1**, 27-64.

Hyvarinen, A. and Oja, E. (2000). Independent Component Analysis: Algorithms and Applications. *Neural Networks*, **13**, 411-430.